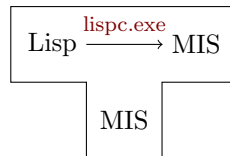


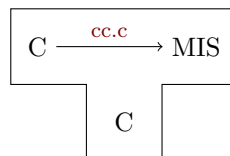
COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 4

2 Compiler Construction (jdy22)

You have acquired an old computer, and you would like to write programs for it in C. Unfortunately, the computer has no executable C compiler: it has only a Lisp compiler `lispc.exe` (an executable that uses the machine's instruction set MIS):

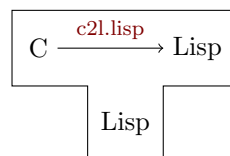


and the source code for a C compiler `cc.c` (written in C):



bootstrapping

- (a) You would like to get the C compiler running efficiently on your computer with as little programming as possible. A useful first step is to write a quick-and-dirty C-to-Lisp compiler in Lisp:

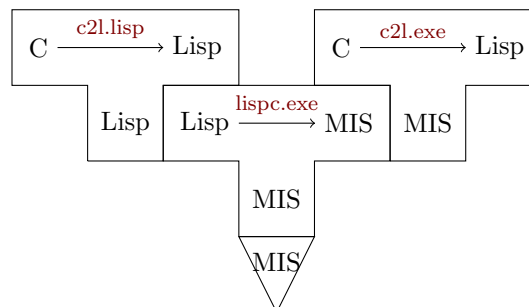


What further steps are needed? Describe and illustrate each step using the same illustration scheme as above.

[8 marks]

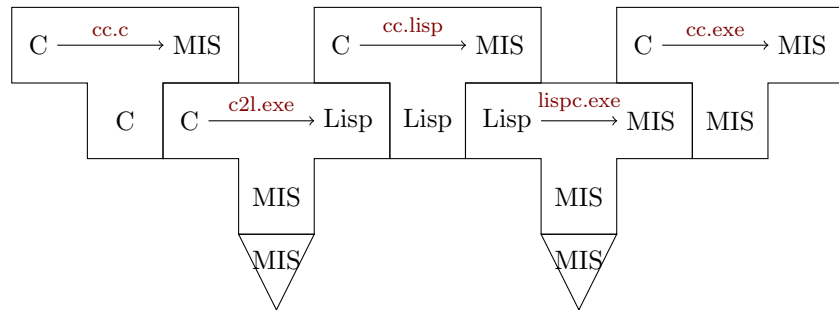
Answer:

- (i) Compile `c2l.lisp` using `lispc.exe` to produce `c2l.exe` [2 marks]

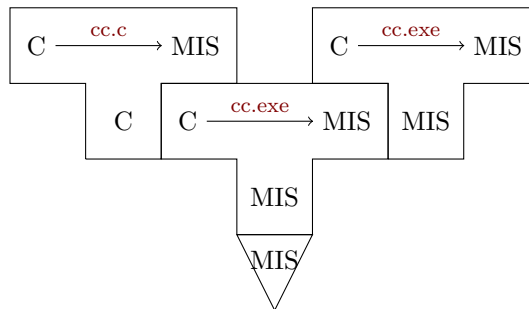


- (ii) Run `c2l.exe` on `cc.c` to produce `cc.lisp` [2 marks], and compile `cc.lisp` using `lispc.exe` to produce `cc.exe` [2 marks]

— *Solution notes* —



(iii) Compile `cc.c` using `cc.exe` [2 marks]



This last step is likely to produce a more efficient compiler, since the code generated directly by `cc.c` is probably better than the code generated by first translating to Lisp.

(b) You would like to extend the C compiler to support foreign function calls so that your C programs can use libraries compiled by `lisp.c.exe`.

(i) What information from the *Application Binary Interface* might you need to ensure that foreign calls work correctly? Comment on function calls, on the stack, on object layout, and on any other relevant considerations.

[6 marks]

Answer: You will need to know about:

- *Passing arguments to functions.* Are arguments passed on the stack or in registers? If they're passed on the stack, in what order are they passed? if they're passed in registers, which registers are used? Similarly, how are values returned from function calls?
- *How the stack is laid out.* What is the format of a stack frame? Does the caller or callee clean up after a call?
- *How objects are laid out in memory.* For example, are there alignment requirements for objects that require padding to be inserted between fields?
- (If you are not using a system linker:) *The format of library files.* For example, where is the symbol table to be found?

This is not exhaustive; answers might also reasonably mention (e.g.) garbage collection. One mark will be awarded per relevant point.

(ii) Programs written in Lisp use a garbage collector (GC) to reclaim unreachable objects. Explain how the GC determines reachability, how

calls between C and Lisp might affect the reachability check, and what changes are needed to make the check work in the presence of those calls.

[6 marks]

Answer: The GC classifies an object as reachable if the object can be reached by transitively following references from the *root set*: pointers held in registers, on the stack, and in global variables [2 marks].

The GC may not include pointers in C stack frames or C data structures in its root set, leading it to incorrectly classify the pointed-to objects as unreachable and reclaim them, and ultimately leading to memory corruption when the reclaimed memory is reused [2 marks].

One way to address the problem is to maintain a global Lisp list of objects passed to C, adding objects to the list when they are passed to C functions and removing them when they are no longer reachable from C [2 marks].
